Wenbo Lyu

B.Sc. Graduate

lyu.geosocial@gmail.com
SpatLyu
wenbo-lyu-824851403
NnaUl34AAAAJ
0009-0002-6003-3800

Curriculum Vitae
September 2026

Education and qualifications

B.Sc. In Geographic Information Science   Shaanxi Normal University   2025

Research Experience

2026.6-	Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6	Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8	Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Selected research papers

1. Lyu, W, Dai, S, Song, Y, Zhao, W, Yi, W, Xiao, Y, and Jia, N (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23. DOI: 10.1080/13658816.2026.2687121.
2. Lyu, W, Lei, Y, Yi, W, Song, Y, Li, X, Dai, S, Qin, Y, and Zhao, W (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems* **127**, 102435. DOI: 10.1016/j.compenvurbsys.2026.102435.
3. Lv, W, Lei, Y, Liu, F, Yan, J, Song, Y, and Zhao, W (2025). gdverse: An R Package for Spatial Stratified Heterogeneity Family. *Transactions in GIS* **29**(2), e70032. DOI: 10.1111/tgis.70032.
4. Lv, W, Liu, F, Cai, K, Cao, Y, Deng, M, Liang, W, Yan, J, and Wang, G (2024). Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China. *Journal of Geophysical Research: Biogeosciences* **129**(10), e2024JG008297. DOI: 10.1029/2024jg008297.
5. Li, B, Lv, W, Ding, H, Song, Y, and Zhao, W (2026). New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China. *GIScience & Remote Sensing* **63**(1), 2716413. DOI: 10.1080/15481603.2026.2716413.
6. Xu, L, Shi, H, Lyu, W, Zhang, Y, and Zhao, H (2026). Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling. *Journal of Transport Geography* **136**, 104789. DOI: 10.1016/j.jtrangeo.2026.104789.
7. Chen, C, Song, Y, Lv, W, Shemery, A, Hampson, K, Yi, W, Zhong, Y, and Wu, P (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. DOI: 10.1111/mice.13489.

Note: Publications under “Lyu, W.” and “Lv, W.” both refer to Wenbo Lyu.